Arista Access Node HEB-ACC-11 QoS Congestion Event Resolution SOP

Node: HEB-ACC-11 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_2883

Description

The QoS congestion event failure domain on Arista access nodes is primarily caused by microbursts, interface tail drops, and QoS queue starvation. This is due to the node's role in handling a large number of traffic flows, which can lead to excessive queue buildup and subsequent congestion. Arista's egress queuing architecture relies on the use of hardware-based queue management, where the TCAM (Ternary Content Addressable Memory) is utilized for packet classification and scheduling. However, in certain scenarios, the TCAM may become overwhelmed, resulting in queue starvation and subsequent tail drops. This can be further exacerbated by the presence of microbursts, which are transient, high-intensity traffic spikes that can overwhelm the node's queuing capacity. To mitigate this issue, it is essential to closely monitor QoS metrics, ensure proper configuration, and implement necessary changes to prevent congestion events.

Pre-requisite Checks

- Pre-requisite: Verify Hebbal Edge site accessibility and ensure remote hands are dispatched for physical intervention if required.

- Pre-requisite: Check the status of redundant links to ensure high availability.

- Pre-requisite: Verify the firmware and software versions of the Arista node are up-to-date.

Potential Root Causes

Cause	Probability
Microbursts and interface tail drops	High
QoS queue starvation	Medium
Inadequate QoS configuration	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware queue, Expected Output: Queuing information for all interfaces, Description: This command displays queuing information for all interfaces on the node, including the number of packets in each queue, the number of packets dropped, and the number of packets transmitted. This information can help identify potential queuing issues and pinpoint the source of congestion events.

Command: show qos statistics, Expected Output: QoS statistics for all interfaces, Description: This command displays QoS statistics for all interfaces on the node, including the number of packets classified, the number of packets transmitted, and the number of packets dropped. This information can help identify potential QoS configuration issues
```

```
and pinpoint the source of congestion events.

Command: show interface detail, Expected Output: Detailed interface information, Description: This command displays detailed information about each interface on the node, including the current speed, duplex, and operational status. This information can help identify potential interface-related issues and pinpoint the source of congestion events.
```

Resolution Steps

Step 1: Step: Reset the egress queues, Command: clear hardware queue, Description: This command resets the egress queues on all interfaces, clearing any accumulated packets and returning the queues to their default state. This can help alleviate congestion events caused by queue buildup.

Step 2: Step: Adjust the QoS configuration, Command: configure terminal set qos policing profile , Description: This command adjusts the QoS configuration to ensure proper policing and shaping of traffic flows, preventing excessive queue buildup and subsequent congestion events.

Step 3: Step: Verify the TCAM configuration, Command: show hardware tcam, Description: This command verifies the TCAM configuration to ensure it is properly set up to handle the current traffic loads. Any issues with the TCAM configuration may contribute to congestion events.

Post-Check Verification

```
Command: show qos statistics, Expected Output: QoS statistics for all interfaces, Description: This command verifies that the QoS configuration is properly applied and functioning as expected.

Command: show hardware queue, Expected Output: Queuing information for all interfaces, Description: This command verifies that the egress queues are operating within expected parameters and that congestion events are no longer occurring.
```

Roll-back Procedures

- Step: Restore the original QoS configuration, Command: configure terminal show running-config | include qos, Description: This command restores the original QoS configuration, ensuring that any changes made during the resolution process are reversed.

- Step: Reset the egress queues, Command: clear hardware queue, Description: This command resets the egress queues to their default state, clearing any accumulated packets.